MIL-C-60260A(MU)
18 June 1970
SUPERSEDING
MIL-C-60260(MU)
22 July 1966

MILITARY SPECIFICATION

CHEST, RANGE FINDER: 10541045

1. SCOPE

1.1 This specification covers a hinged, watertight chest having a displacement of approximately 2.7 cubic feet and hardware to secure the chest to a vehicle (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10541026	Case, Carrying
F10541038	Case, Carrying
F10541045	Chest, Range Finder: 10541045

FSC 1240

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

PACKAGING DATA SHEET

10541045 Packaging of Chest, Range Finder: 10541045

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The chest shall be manufactured in accordance with Drawing F10541045 and drawings pertaining thereto.

3.3 General specification.- The requirements of MIL-F-13926 listed below shall be met wherever applicable:

a. Dimensions and tolerances.
b. Inorganic protective surface finishes.
c. Part identification and marking.

3.4 Workmanship.- The workmanship shall meet the requirements of MIL-F-13926. The protective surface finishes shall be free of voids which may affect their protective qualitites or appearance.

3.5 Storage temperature.- The chest shall show no evidence of physical damage when stored at temperatures from -80° to +160°F with its contents positioned within the chest and the cover securely latched to the body of the chest. Subsequent to exposure to the temperatures specified herein, the chest shall meet the requirements of 3.8 at temperatures between 70° and +80°F and shall also meet the requirements of 3.9 and 3.10.

3.6 Shock.- When loaded with its contents and secured by the mounting hardware the chest and mounting hardware shall show no evidence of physical damage when subjected to shock impulse in each of 3 mutually perpendicular directions. Each shock impulse shall be a half sine wave having a time duration of 11 ± 1 milliseconds and a peak amplitude of 75g. Subsequent to shock, the chest shall meet the requirements of 3.8 at a temperature between +70 and +80°F and shall also meet the requirements of 3.9 and 3.10.

3.7 Vibration.- When loaded with its contents and secured by the mounting hardware, the chest shall show no evidence of physical failure due to being vibrated for 15 minutes in each of 3 mutually perpendicular directions. The frequency shall be varied continuously between 10 and 55 cycles per second (cps) at a rate not exceeding 1.0 cps. The double amplitude shall be 0.031 inch. Subsequent to vibration, the chest shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

meet the requirements of 3.8 at temperatures between +70° and +80°F and shall also meet the requirements of 3.9 and 3.10.

3.8 Operability of catches, hinges and pressure relief valves.- While the chest is thermally stabilized at temperatures of -65°F and +155°F and at all temperatures between -65°F and +155°F, the catches, hinges and pressure relief valves shall be manually operable.

3.9 Immersion.- The chest shall show no evidence of leakage when immersed under a 14 to 16 inch head of water for a period of at least 30 minutes. The temperature of the water shall be between +70° and +80°F.

3.10 Interface requirements.- The chest shall interface, support and retain:

a. Carrying Case, 10541026 and contents.
b. Carrying Case, 10541038 and contents.

The chest shall retain the intended contents (see 6.1) and allow no free movement of the contents when the cover is closed and securely latched to the body of the chest. The chest shall mount to a vehicle using the channels on the bottom of the chest in conjunction with the mounting hardware.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is reponsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirementsof the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% DEFECTIVE		
101. Operability of catches, hinges and pressure relief valves	3.8	4.6.4
102. Immersion	3.9	4.6.5
103. Interface requirements	3.10	4.6.6
MINOR: NONE		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three chests shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable dwgs & specs - visual
302.	Fabrication	3.2	Applicable dwgs & specs - visual
303.	General specification	3.3	MIL-F-13926 - visual
304.	Workmanship	3.4	MIL-F-13926

NOTE: The tests in Table II shall be conducted at a temperature between +70°F and +80°F.

4.4.2 Environmental.- Three chests shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
305.	Vibration (+70°F to +80°F)	3.7	4.6.1
306.	Shock (+70° to +80°F)	3.6	4.6.2
307.	Storage temperatures (-80°F to +160°F)	3.5	4.6.3

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3.).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.- During each environmental test listed in Table III, the chest shall contain 2 simulated loads as specified in Figure I, Drawing F10541038 and drawings pertaining thereto. The simulated loads shall be to the dimensions and weight shown on Figure I. The chest shall be secured to a plywood (at least 1" thick) base by means of mounting hardware as shown on Drawing 10541045. The sequence of environmental tests shall be as follows:

a.	Vibration	4.6.1
b.	Shock	4.6.2
c.	Storage temperature	4.6.3

4.6.1 Vibration.- The test equipment utilized in this test shall conform to the vibration test equipment requirements specified in MIL-STD-810. The chest shall contain the simulated loads and secured by mounting hardware to a plywood base which is secured to the vibration table. The vibration equipment shall be capable of providing the conditions of acceleration amplitude, frequency and duration specified in 3.7. The vibration test shall be performed in accordance with Test Method 514 of MIL-STD-810. Upon completion of the vibration test, the chest shall be subjected to the tests specified in 4.6.4 and 4.6.6.

4.6.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The chest shall contain 2 simulated loads and secured by the mounting hardware

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

to a plywood base. The test shall be performed as specified in Procedure II of MIL-F-13926, except the shock pulses shall comply with the wave form, time duration and peak "G" value specified in 3.6. Upon completion of the shock test the chest shall be subjected to the tests specified in 4.6.4 and 4.6.6.

4.6.3 Storage temperature.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 with the following exceptions. After the chest has stabilized at -80°F for at least 2 hours, increase the temperature to -65°F ± 3°F and stabilized for 1 hour. After the chest is stabilized at -65°F ± 3°F, the chest shall be subjected to test specified in 4.6.4. The temperature of the chamber then shall be stabilized at +70°F ± 5°F for 1 hour and the chest subjected to the test specified in 4.6.4. The temperature of the chamber then shall be increased to +160°F ± 3°F and stabilized for 2 hours, after which the chamber temperature shall be reduced to +155°F ± 3°F and stabilized for 1 hour. The chest shall again be tested as specified in 4.6.4. The temperature shall be reduced to 70°F ± 3°F and the chest shall meet the requirements of 4.6.4, 4.6.5 and 4.6.6 of this specification.

4.6.4 Operability of catches, hinges and pressure relief valves.- The catches and hinges shall be visually examined for damage and shall be capable of movement without binding. Using representative configuration of the carrying cases (simulated loads) as shown in Figure I, chest inserts shall support cases at contacting interfaces, chest cover shall close within 3 inches and then by applying a maximum load of 50 lbs. to the cover, it shall close and be capable of being secured with the catches. The pressure relief valves shall meet the requirements specified in 3.8.

4.6.5 Immersion test.- The test equipment utilized in this test shall be in accordance with the test facilities requirements of MIL-F-13926. The chest shall contain simulated loads as shown in Figure I. The locked case shall be exposed to the requirements of 3.9, after which the interior of the chest shall show no evidence of leakage or moisture. This test shall be performed after the vibration, shock, and storage temperature tests have been completed.

4.6.6 Interface requirements.- This test is to be performed using the 2 simulated loads shown on Figure I. All interfaces shall be visually inspected to insure proper fit of simulated loads and to provide protection of chest contents undergoing test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10541045. The level of protection shall be as specified in the procurement document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use.- The Range Finder Chest, 10541045 is primarily intended to house 2 carrying cases with contents, a training manual, an equipment log book and a boresight chart. Carrying Case 10541038 will contain the range finder. Carrying Case 10541026 will contain accessory equipment for the range finder. This accessory equipment includes a tripod, adapter mount with cable, battery pack, and shelf mount. A document compartment is provided for storing the technical manual, log book, and boresight chart.

6.2 Ordering data.- Procurement documents should specify:

a. Title, date and number of this specification.
b. Applicable packaging data sheet.
c. Level of protection required.
d. Provisions of first article testing.

Custodian:

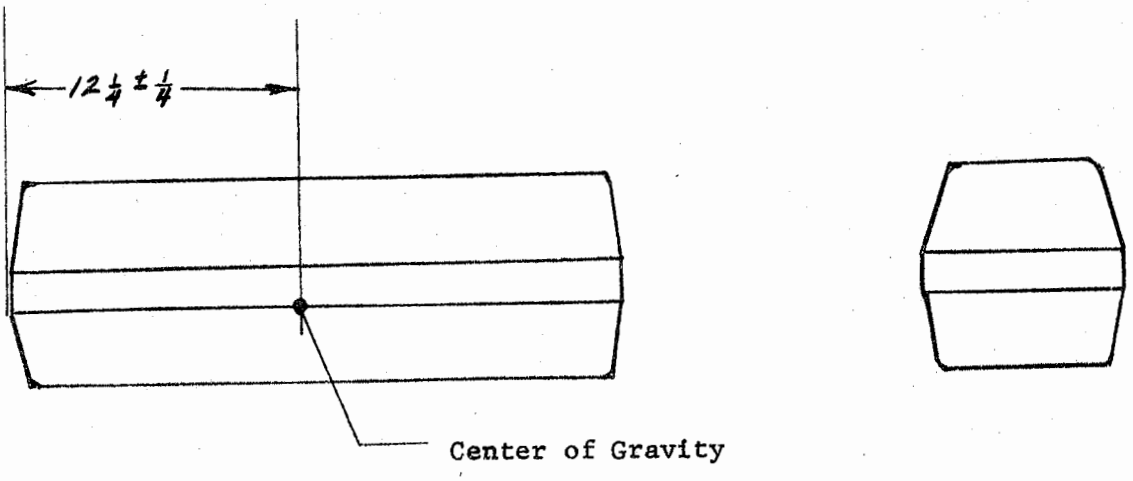
Army - MU

Preparing activity:

Army - MU

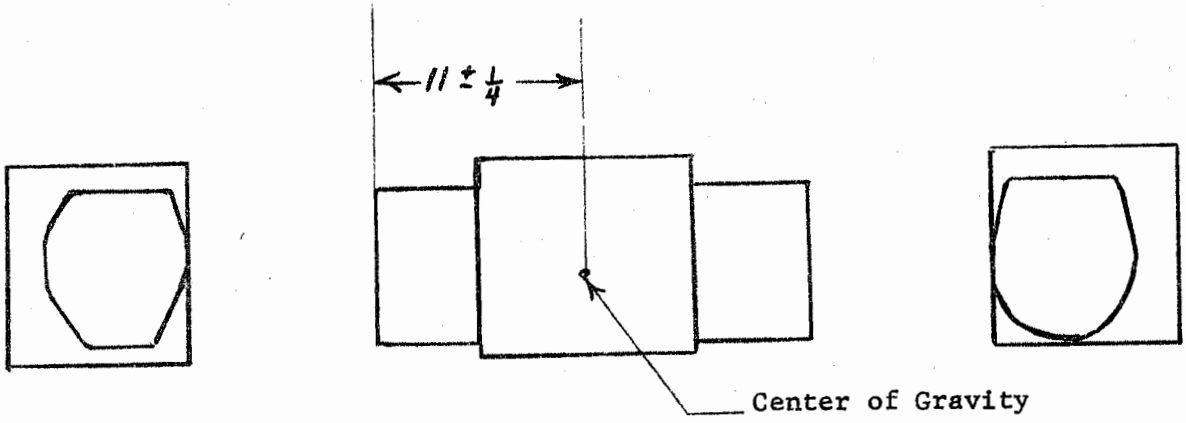
Project No. 1240-A221

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.



)TES:
1. Refer to Drawing F10541026 and drawings pertaining thereto, for all other nece
2. Weight: 14 1/4 lbs. ± 1/4 lb dimensi

Simulated Carrying Case 10541026 Load



:
Refer to Drawing F10541038 and drawings pertaining thereto for all other necessa dimensions.

Weight 21 1/2 lbs. ± 1/4 lb.

Simulated Carrying Case 10541038 Load

FIGURE I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:24Z
Check the source to verify that this is the current version before use.